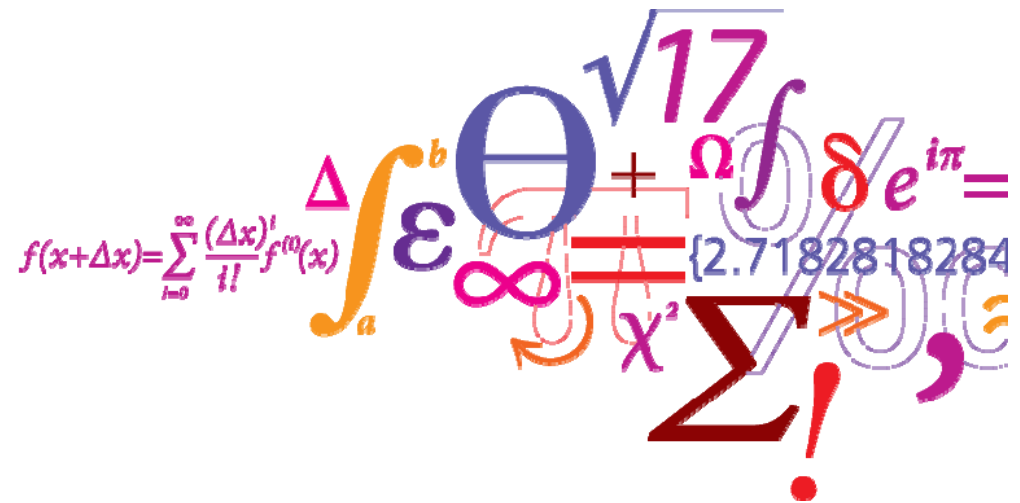


31792 -- Advanced Optimization and Game Theory for Energy Systems

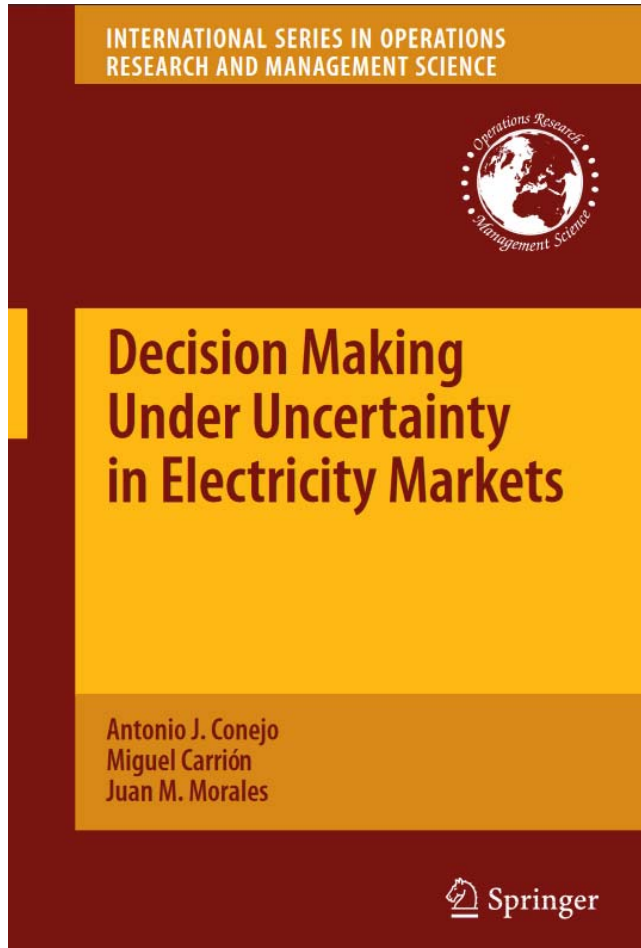
Lecture 5: Stochastic market clearing

Jalal Kazempour

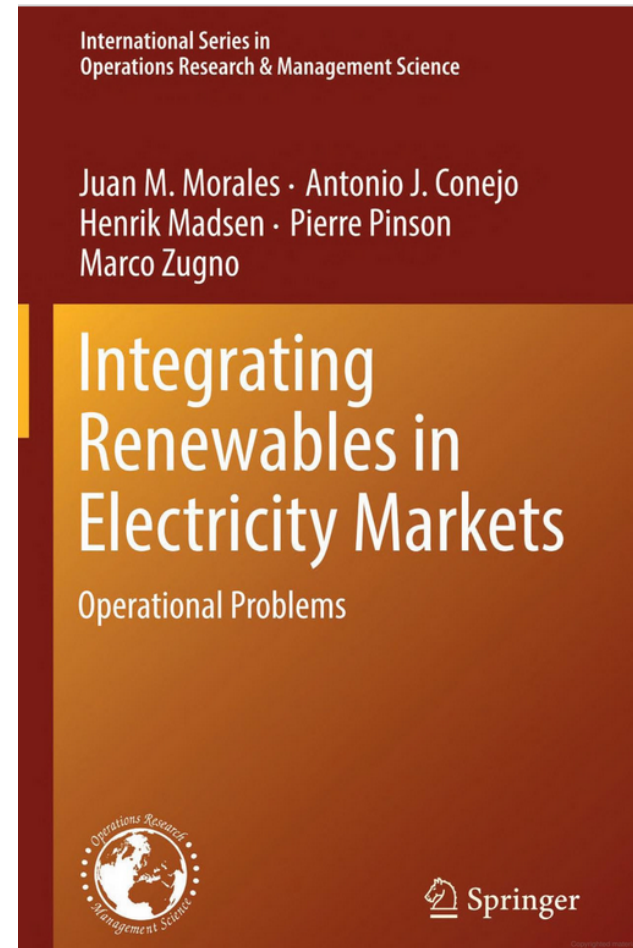
January 11, 2021



References

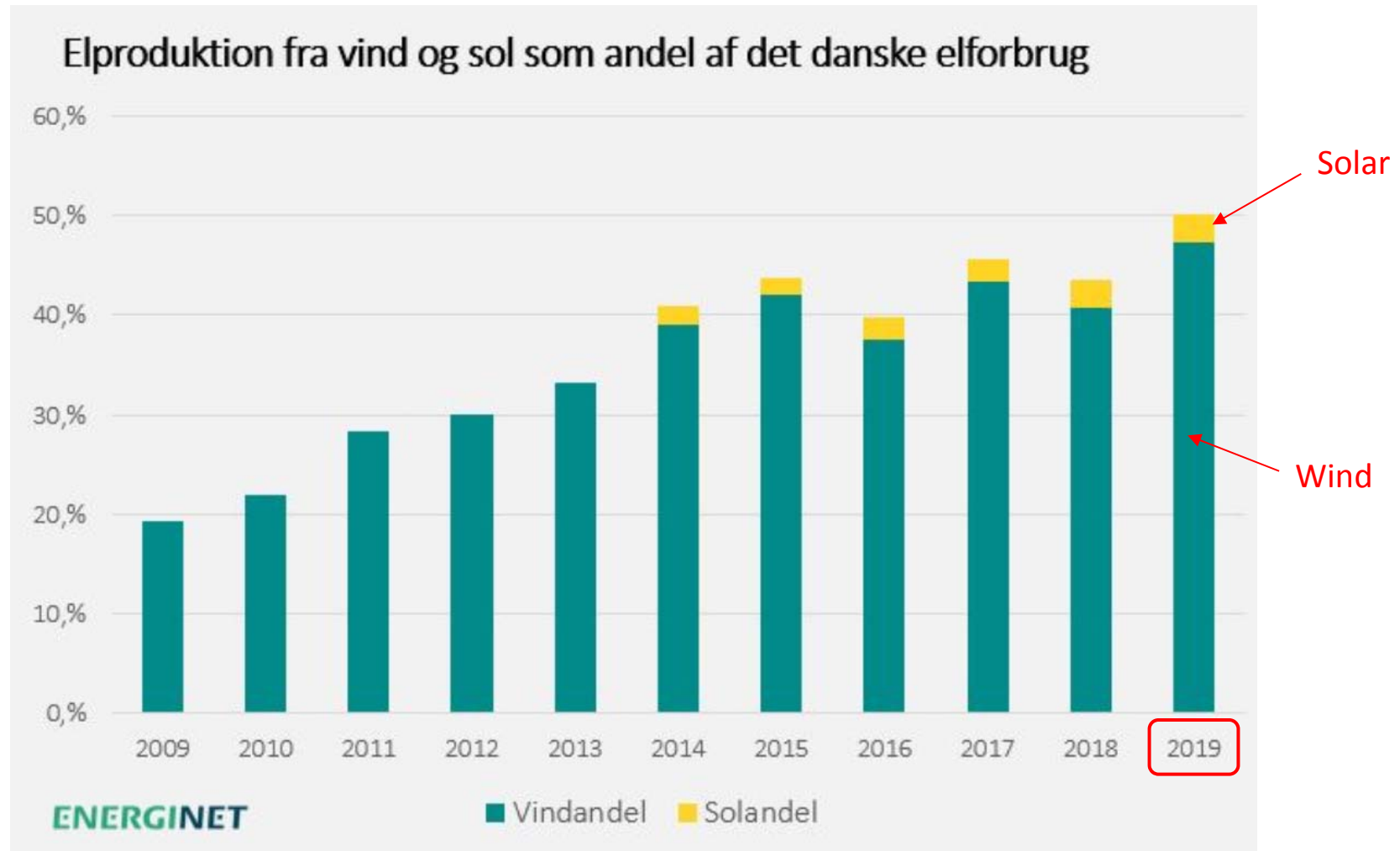


DTU Electrical Engineering, Technical University of Denmark



Jalal Kazempour

This is happening in Denmark!



Uncertainty as a part of market!

Electricity Market

Goal: meeting demand at the maximum social welfare
(or the minimum total operational cost)

Uncertainty as a part of market!

Electricity Market

Goal: meeting demand at the maximum social welfare
(or the minimum total operational cost)

- **Renewables** (with stochastic generation) bring uncertainty: inaccurate forecast may result in wrong commitment and dispatch decisions, with increased total operational cost for the system

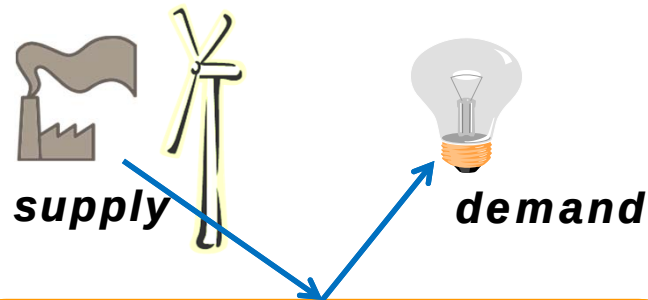
Uncertainty as a part of market!

Electricity Market

Goal: meeting demand at the maximum social welfare
(or the minimum total operational cost)

- **Renewables** (with stochastic generation) bring uncertainty: inaccurate forecast may result in wrong commitment and dispatch decisions, with increased total operational cost for the system
- **How to manage renewable power uncertainty:**
 - ✓ Flexibility integration (fast generators, storage, demand response, etc)
 - ✓ Proper market design

Two-Stage Settlement, 1 Day Horizon

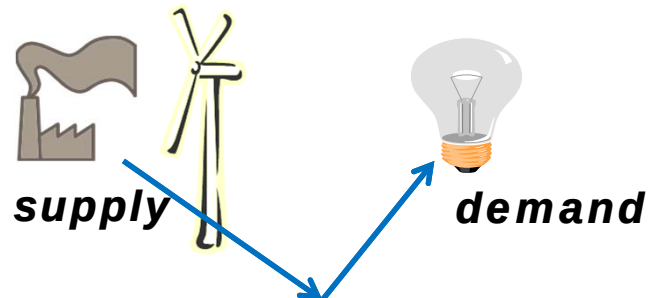


Day-ahead (DA) Market

(Operator balances supply and demand, using a forecast of load and wind)

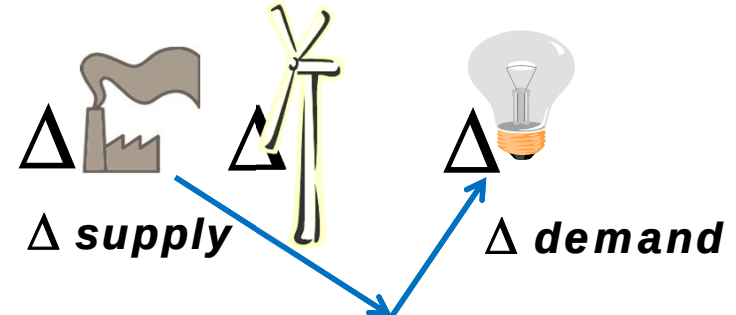
To be cleared 24-36 hours before the real time!

Two-Stage Settlement, 1 Day Horizon



Day-ahead (DA) Market

(Operator balances supply and demand, using a forecast of load and wind)



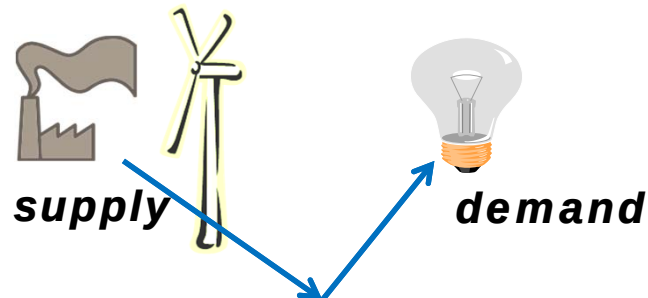
Real-time (RT) Market

(Operator clears imbalances using actual load and wind)



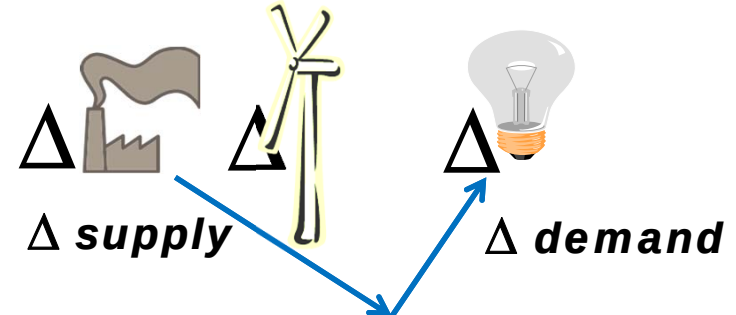
To be cleared 24-36 hours before the real time!

Two-Stage Settlement, 1 Day Horizon



Day-ahead (DA) Market

(Operator balances supply and demand, using a forecast of load and wind)



Real-time (RT) Market

(Operator clears imbalances using actual load and wind)



To be cleared 24-36 hours before the real time!

Note: There are more short-run settlements in the real-world electricity markets, e.g., intra-day markets! For simplicity, let's ignore them!

Illustrative example



- Capacity: 100 MW
- Production cost: \$10/MWh
- **Inflexible** in real time (RT)



- Capacity: 70 MW
- Production cost: \$0/MWh
- Uncertain production in DA
- Realization in RT: 30 MW



- Capacity: 30 MW
- Production cost: \$20/MWh
- **Fully flexible** in RT



- 120 MW (inelastic and deterministic)
- Curtailment cost: \$80/MWh

Illustrative example



- Capacity: 100 MW
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- Inflexible in real time (RT)



- Capacity: 30 MW
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Wind forecast in DA [MW]			
30 (perfect forecast)			

Illustrative example



- Capacity: 100 MW
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Load

- 120 MW (inelastic and deterministic)
- Curtailment cost: \$80/MWh

Wind forecast in DA [MW]	DA market outcomes		
30 (perfect forecast)	WP: ? G1: ? G2: ? Cost ^{DA} : ?		

Illustrative example



- Capacity: 100 MW
- Production cost: \$10/MWh
- **Inflexible** in real time (RT)



- Capacity: 30 MW
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- Capacity: 70 MW
- Production cost: \$0/MWh
- Uncertain production in DA
- Realization in RT: 30 MW



Load

- 120 MW (inelastic and deterministic)
- Curtailment cost: \$80/MWh

Wind forecast in DA [MW]	DA market outcomes		
30 (perfect forecast)	WP: 30 MW G1: 90 MW G2: 0 MW Cost ^{DA} : \$900		

Illustrative example



- Capacity: 100 MW
- Production cost: \$10/MWh
- **Inflexible** in real time (RT)



- Capacity: 30 MW
- Production cost: \$20/MWh
- **Fully flexible** in RT



- Capacity: 70 MW
- Production cost: \$0/MWh
- Uncertain production in DA
- Realization in RT: 30 MW



Load

- 120 MW (inelastic and deterministic)
- Curtailment cost: \$80/MWh

Wind forecast in DA [MW]	DA market outcomes	RT market outcomes	
30 (perfect forecast)	WP: 30 MW G1: 90 MW G2: 0 MW Cost ^{DA} : \$900	WP: ? G1: ? G2: ? Cost ^{RT} (additional cost): ?	

Illustrative example



- Capacity: 100 MW
- Production cost: \$10/MWh
- **Inflexible** in real time (RT)



- Capacity: 30 MW
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- Capacity: 70 MW
- Production cost: \$0/MWh
- Uncertain production in DA
- Realization in RT: 30 MW



Load

- 120 MW (inelastic and deterministic)
- Curtailment cost: \$80/MWh

Wind forecast in DA [MW]	DA market outcomes	RT market outcomes	
30 (perfect forecast)	WP: 30 MW G1: 90 MW G2: 0 MW Cost ^{DA} : \$900	WP: unchanged G1: unchanged G2: unchanged Cost ^{RT} (additional cost): \$0	

Illustrative example



- Capacity: 100 MW
- Production cost: \$10/MWh
- **Inflexible** in real time (RT)



- Capacity: 30 MW
- Production cost: \$20/MWh
- **Fully flexible** in RT



- Capacity: 70 MW
- Production cost: \$0/MWh
- Uncertain production in DA
- Realization in RT: 30 MW



Load

- 120 MW (inelastic and deterministic)
- Curtailment cost: \$80/MWh

Wind forecast in DA [MW]	DA market outcomes	RT market outcomes	Total operational cost in DA and RT [\$]
30 (perfect forecast)	WP: 30 MW G1: 90 MW G2: 0 MW Cost ^{DA} : \$900	WP: unchanged G1: unchanged G2: unchanged Cost ^{RT} (additional cost): \$0	?

Illustrative example



- Capacity: 100 MW
- Production cost: \$10/MWh
- **Inflexible** in real time (RT)



- Capacity: 30 MW
- Production cost: \$20/MWh
- **Fully flexible** in RT



- Capacity: 70 MW
- Production cost: \$0/MWh
- Uncertain production in DA
- Realization in RT: 30 MW



Load

- 120 MW (inelastic and deterministic)
- Curtailment cost: \$80/MWh

Wind forecast in DA [MW]	DA market outcomes	RT market outcomes	Total operational cost in DA and RT [\$]
30 (perfect forecast)	WP: 30 MW G1: 90 MW G2: 0 MW Cost ^{DA} : \$900	WP: unchanged G1: unchanged G2: unchanged Cost ^{RT} (additional cost): \$0	Cost ^{DA} + Cost ^{RT} = 900

Illustrative example



- Capacity: 100 MW
- Production cost: \$10/MWh
- **Inflexible** in real time (RT)



- Capacity: 30 MW
- Production cost: \$20/MWh
- **Fully flexible** in RT



- Capacity: 70 MW
- Production cost: \$0/MWh
- Uncertain production in DA
- Realization in RT: 30 MW



Load

- 120 MW (inelastic and deterministic)
- Curtailment cost: \$80/MWh

Despite all advances in forecasting methods, the wind forecast can still be erroneous!

Let's now consider an *imperfect* wind power forecast in the DA stage!

Illustrative example



- Capacity: 100 MW
- Production cost: \$10/MWh
- **Inflexible** in real time (RT)



- Capacity: 30 MW
- Production cost: \$20/MWh
- **Fully flexible** in RT



- Capacity: 70 MW
- Production cost: \$0/MWh
- Uncertain production in DA
- Realization in RT: 30 MW



- 120 MW (inelastic and deterministic)
- Curtailment cost: \$80/MWh

Wind forecast in DA [MW]	DA market outcomes	RT market outcomes	Total operational cost in DA and RT [\$]
60			

Illustrative example



- Capacity: 100 MW
- Production cost: \$10/MWh
- **Inflexible** in real time (RT)



- Capacity: 30 MW
- Production cost: \$20/MWh
- **Fully flexible** in RT



- Capacity: 70 MW
- Production cost: \$0/MWh
- Uncertain production in DA
- Realization in RT: 30 MW



- 120 MW (inelastic and deterministic)
- Curtailment cost: \$80/MWh

Wind forecast in DA [MW]	DA market outcomes	RT market outcomes	Total operational cost in DA and RT [\$]
--------------------------------	-----------------------	--------------------	--

60 ← The DA forecast is optimistic, and there will be a wind power deficit of 30 MW in RT!

Illustrative example



- Capacity: 100 MW
- Production cost: \$10/MWh
- **Inflexible** in real time (RT)



- Capacity: 30 MW
- Production cost: \$20/MWh
- **Fully flexible** in RT



- Capacity: 70 MW
- Production cost: \$0/MWh
- Uncertain production in DA
- Realization in RT: 30 MW



Load

- 120 MW (inelastic and deterministic)
- Curtailment cost: \$80/MWh

Wind forecast in DA [MW]	DA market outcomes	RT market outcomes	Total operational cost in DA and RT [\$]
60	WP: ? G1: ? G2: ? Cost ^{DA} : ?	WP: ? G1: ? G2: ? Cost ^{RT} : ?	?

Illustrative example



- Capacity: 100 MW
- Production cost: \$10/MWh
- **Inflexible** in real time (RT)



- Capacity: 30 MW
- Production cost: \$20/MWh
- **Fully flexible** in RT



- Capacity: 70 MW
- Production cost: \$0/MWh
- Uncertain production in DA
- Realization in RT: 30 MW



- 120 MW (inelastic and deterministic)
- Curtailment cost: \$80/MWh

Wind forecast in DA [MW]	DA market outcomes	RT market outcomes	Total operational cost in DA and RT [\$]
60	WP: 60 MW G1: 60 MW G2: 0 MW Cost ^{DA} : \$600	WP: ? G1: ? G2: ? Cost ^{RT} : ?	?

Illustrative example



- Capacity: 100 MW
- Production cost: \$10/MWh
- **Inflexible** in real time (RT)



- Capacity: 30 MW
- Production cost: \$20/MWh
- **Fully flexible** in RT



- Capacity: 70 MW
- Production cost: \$0/MWh
- Uncertain production in DA
- Realization in RT: 30 MW



Load

- 120 MW (inelastic and deterministic)
- Curtailment cost: \$80/MWh

Wind forecast in DA [MW]	DA market outcomes	RT market outcomes	Total operational cost in DA and RT [\$]
60	WP: 60 MW G1: 60 MW G2: 0 MW Cost ^{DA} : \$600	WP: -30 MW (production deficit) G1: unchanged G2: +30 MW (up regulation) Cost ^{RT} : \$600	1200

Illustrative example



- Capacity: 100 MW
- Production cost: \$10/MWh
- **Inflexible** in real time (RT)



- Capacity: 30 MW
- Production cost: \$20/MWh
- **Fully flexible** in RT



- Capacity: 70 MW
- Production cost: \$0/MWh
- Uncertain production in DA
- Realization in RT: **30 MW**



- 120 MW (inelastic and deterministic)
- Curtailment cost: \$80/MWh

Wind forecast in DA [MW]	DA market outcomes	RT market outcomes	Total operational cost in DA and RT [\$]
--------------------------------	-----------------------	--------------------	--

70 ← The DA forecast is optimistic, and there will be a wind power **deficit of 40 MW** in RT!

Illustrative example



- Capacity: 100 MW
- Production cost: \$10/MWh
- **Inflexible** in real time (RT)



- Capacity: 30 MW
- Production cost: \$20/MWh
- **Fully flexible** in RT



- Capacity: 70 MW
- Production cost: \$0/MWh
- Uncertain production in DA
- Realization in RT: **30 MW**



Load

- 120 MW (inelastic and deterministic)
- Curtailment cost: \$80/MWh

Wind forecast in DA [MW]	DA market outcomes	RT market outcomes	Total operational cost in DA and RT [\$]
70	WP: 70 MW G1: 50 MW G2: 0 MW Cost ^{DA} : \$500	WP: -40 MW (production deficit) G1: unchanged G2: +30 MW (up regulation) Load curtailed: +10 MW Cost ^{RT} : \$1400	1900

Illustrative example



- Capacity: 100 MW
- Production cost: \$10/MWh
- **Inflexible** in real time (RT)



- Capacity: 30 MW
- Production cost: \$20/MWh
- **Fully flexible** in RT



- Capacity: 70 MW
- Production cost: \$0/MWh
- Uncertain production in DA
- Realization in RT: 30 MW



- 120 MW (inelastic and deterministic)
- Curtailment cost: \$80/MWh

Wind forecast in DA [MW]	DA market outcomes	RT market outcomes	Total operational cost in DA and RT [\$]
--------------------------------	-----------------------	--------------------	--

10 ← The DA forecast is pessimistic, and there will be a wind power **excess of 20 MW** in RT!

Illustrative example



- Capacity: 100 MW
- Production cost: \$10/MWh
- **Inflexible** in real time (RT)



- Capacity: 30 MW
- Production cost: \$20/MWh
- **Fully flexible** in RT



- Capacity: 70 MW
- Production cost: \$0/MWh
- Uncertain production in DA
- Realization in RT: **30 MW**



Load

- 120 MW (inelastic and deterministic)
- Curtailment cost: \$80/MWh

Wind forecast in DA [MW]	DA market outcomes	RT market outcomes	Total operational cost in DA and RT [\$]
10	WP: 10 MW G1: 100 MW G2: 10 MW Cost ^{DA} : \$1200	WP: +20 MW (production excess) G1: unchanged G2: -10 MW (down regulation) Wind spilled: +10 MW Cost ^{RT} : -\$200	1000

Illustrative example

Wind forecast in DA [MW]	DA cost [\$]	RT cost [\$]	Total cost (DA+RT) [\$]
30 (perfect forecast)	900	0	900
60	600	600	1200
70	500	1400	1900
10	1200	-200	1000
Average forecast $= [30 + 60 + 70 + 10] / 4$ $= 42.5 \text{ MW}$	775	250	1025

Discussion

- Recall that we don't know the actual wind power realization in DA.
- We only have a wind forecast in DA!

Discussion

- Recall that we don't know the actual wind power realization in DA.
- We only have a wind forecast in DA!
- Which forecast should be considered in DA?
 - 30 MW?
 - 60 MW?
 - 70 MW?
 - 10 MW?
 - Average forecast?
 - Any other value?



Discussion

- In DA stage:

Isn't it smarter to consider each wind forecast as a potential "scenario" in RT, evaluate market consequences under every RT scenario, and then determine DA dispatch according to all scenarios?



Discussion

- In DA stage:

Isn't it smarter to consider each wind forecast as a potential "scenario" in RT, evaluate market consequences under every RT scenario, and then determine DA dispatch according to all scenarios?



RT stage

- > RT scenario 1: Wind power is 30 MW
- > RT scenario 2: Wind power is 60 MW
- > RT scenario 3: Wind power is 70 MW
- > RT scenario 4: Wind power is 10 MW

Discussion

- In DA stage:

Isn't it smarter to consider each wind forecast as a potential "scenario" in RT, evaluate market consequences under every RT scenario, and then determine DA dispatch according to all scenarios?



DA stage

DA dispatch to be determined based on all RT scenarios!

RT stage

- RT scenario 1: Wind power is 30 MW
- RT scenario 2: Wind power is 60 MW
- RT scenario 3: Wind power is 70 MW
- RT scenario 4: Wind power is 10 MW

Discussion

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Isn't it smarter to consider each wind forecast as a potential "scenario" in RT, evaluate market consequences under every RT scenario, and then determine DA dispatch according to all scenarios?



DA stage

DA dispatch to be determined based on all RT scenarios!

A weight (probability) to be assigned to each scenario!

RT stage

- RT scenario 1: Wind power is 30 MW
- RT scenario 2: Wind power is 60 MW
- RT scenario 3: Wind power is 70 MW
- RT scenario 4: Wind power is 10 MW

Discussion

- In DA stage:

Isn't it smarter to consider each wind forecast as a potential "scenario" in RT, evaluate market consequences under every RT scenario, and then determine DA dispatch according to all scenarios?



DA stage

DA dispatch to be determined based on all RT scenarios!

A weight (probability) to be assigned to each scenario!

RT stage

0.25 → RT scenario 1: Wind power is 30 MW

0.25 → RT scenario 2: Wind power is 60 MW

0.25 → RT scenario 3: Wind power is 70 MW

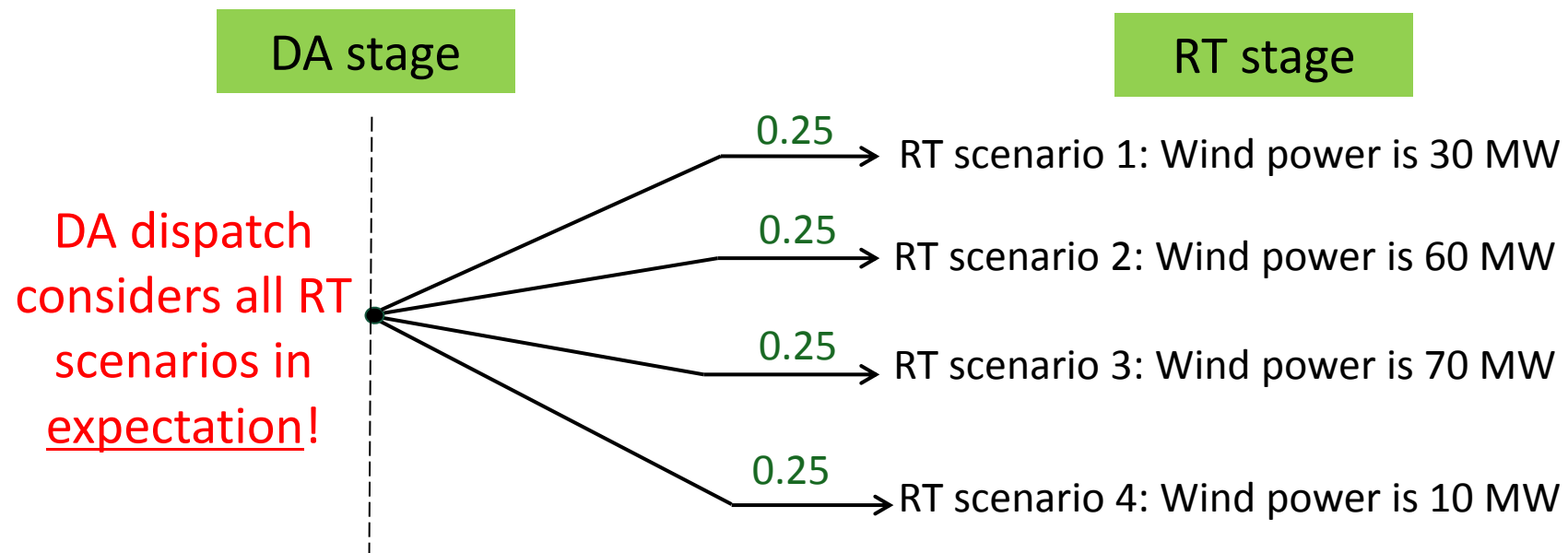
0.25 → RT scenario 4: Wind power is 10 MW

Let's assume four scenarios are equiprobable!

Discussion

- In DA stage:

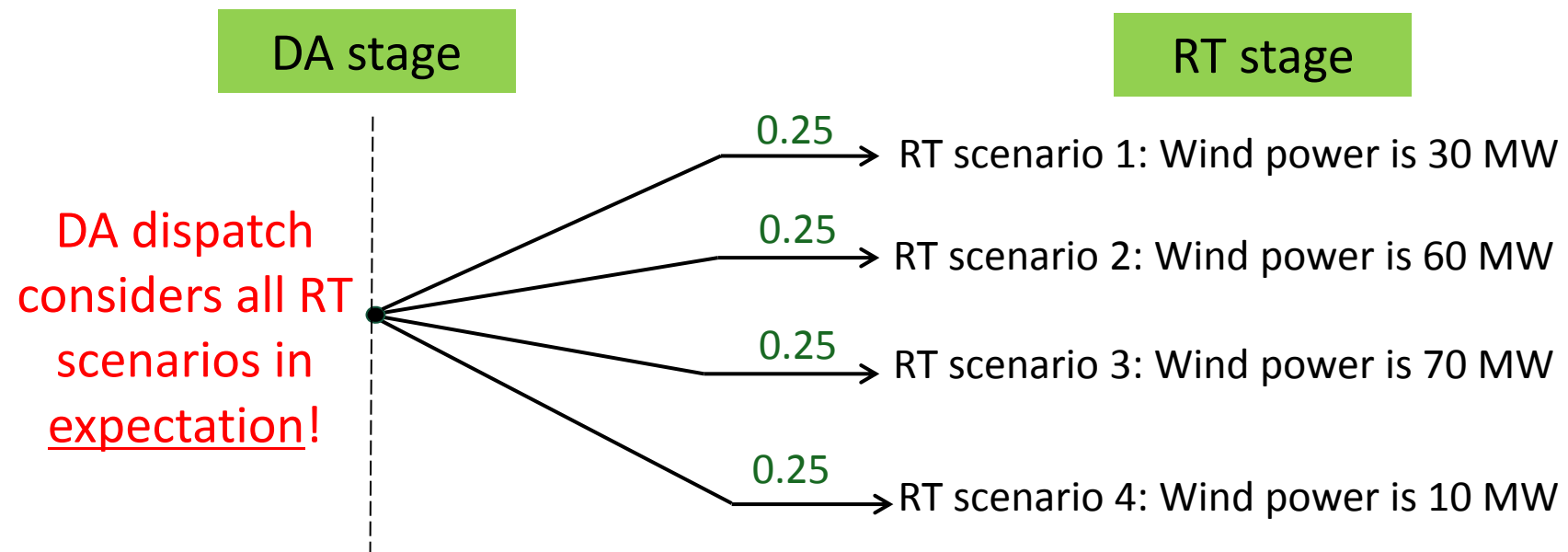
Isn't it smarter to consider each wind forecast as a potential "scenario" in RT, evaluate market consequences under every RT scenario, and then determine DA dispatch according to all scenarios?



Discussion

- In DA stage:

Look-ahead strategy

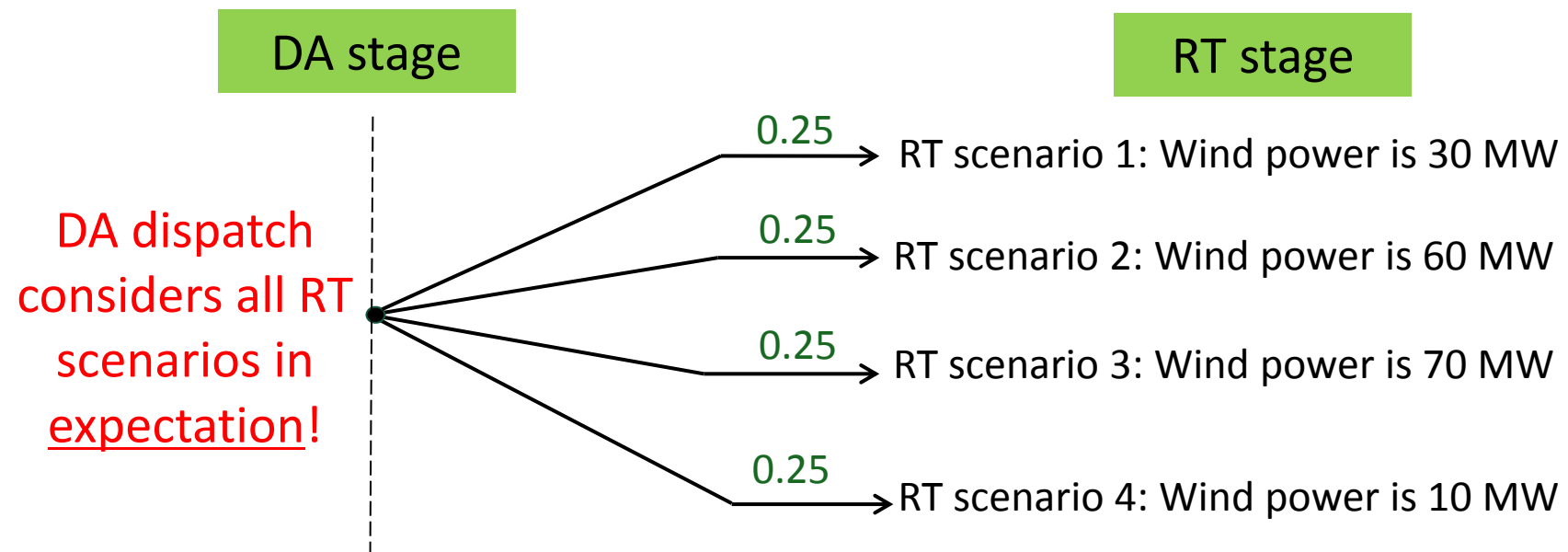


Discussion

- In DA stage:

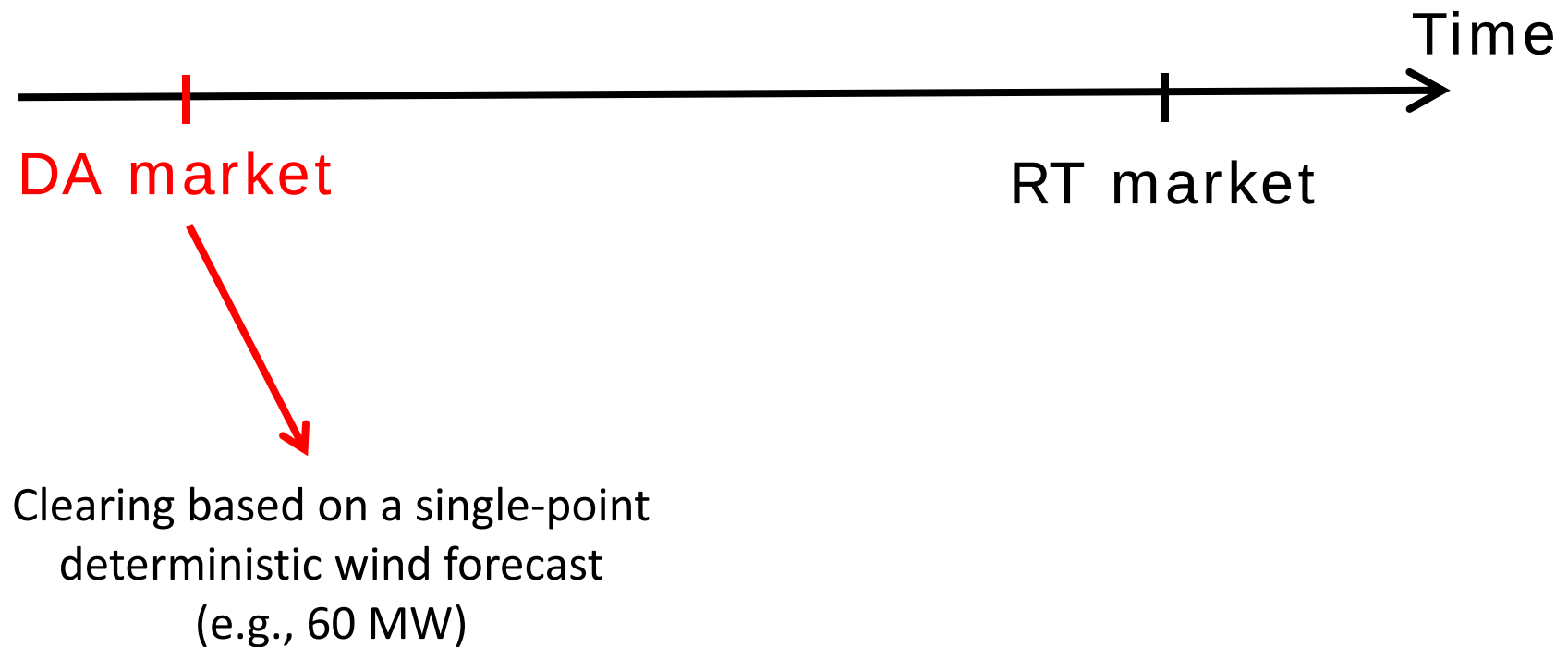
Look-ahead strategy

Stochastic market-clearing mechanism



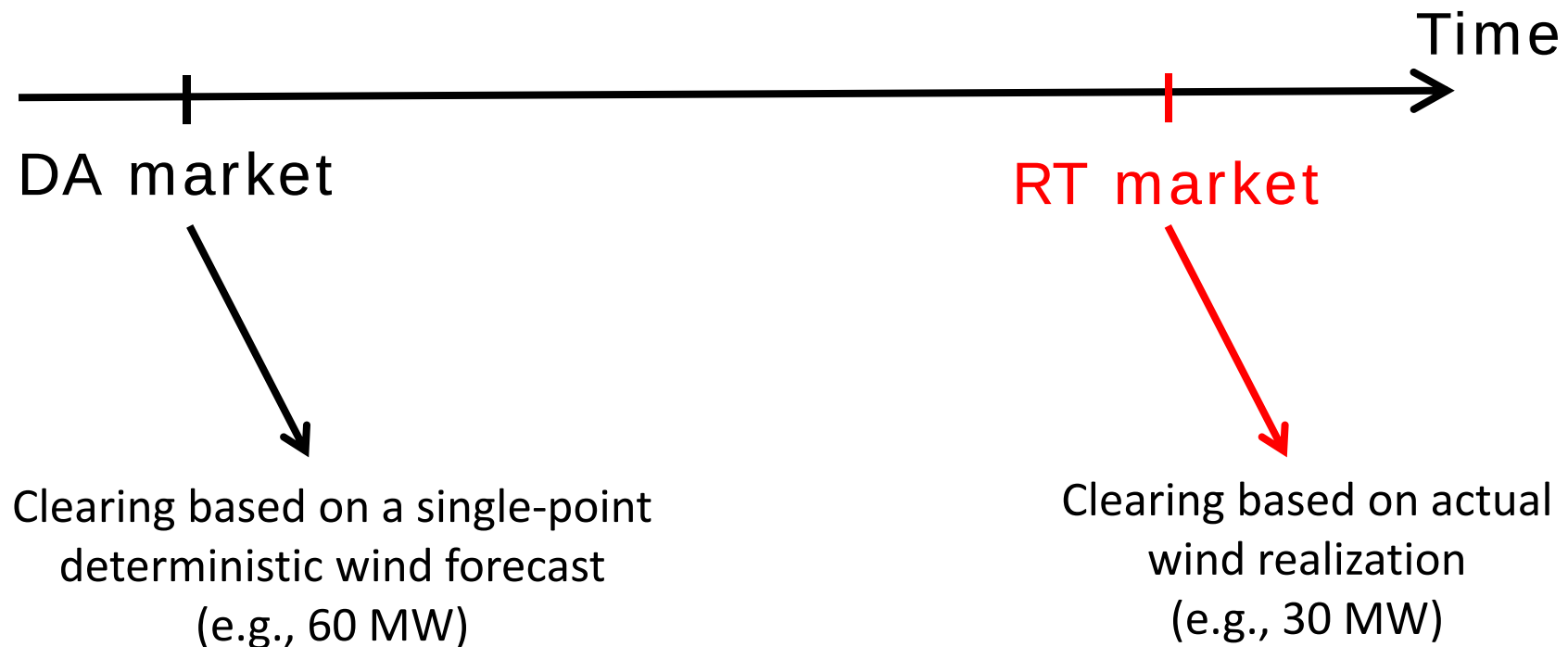
Deterministic vs stochastic market clearing

In a **deterministic** market mechanism:



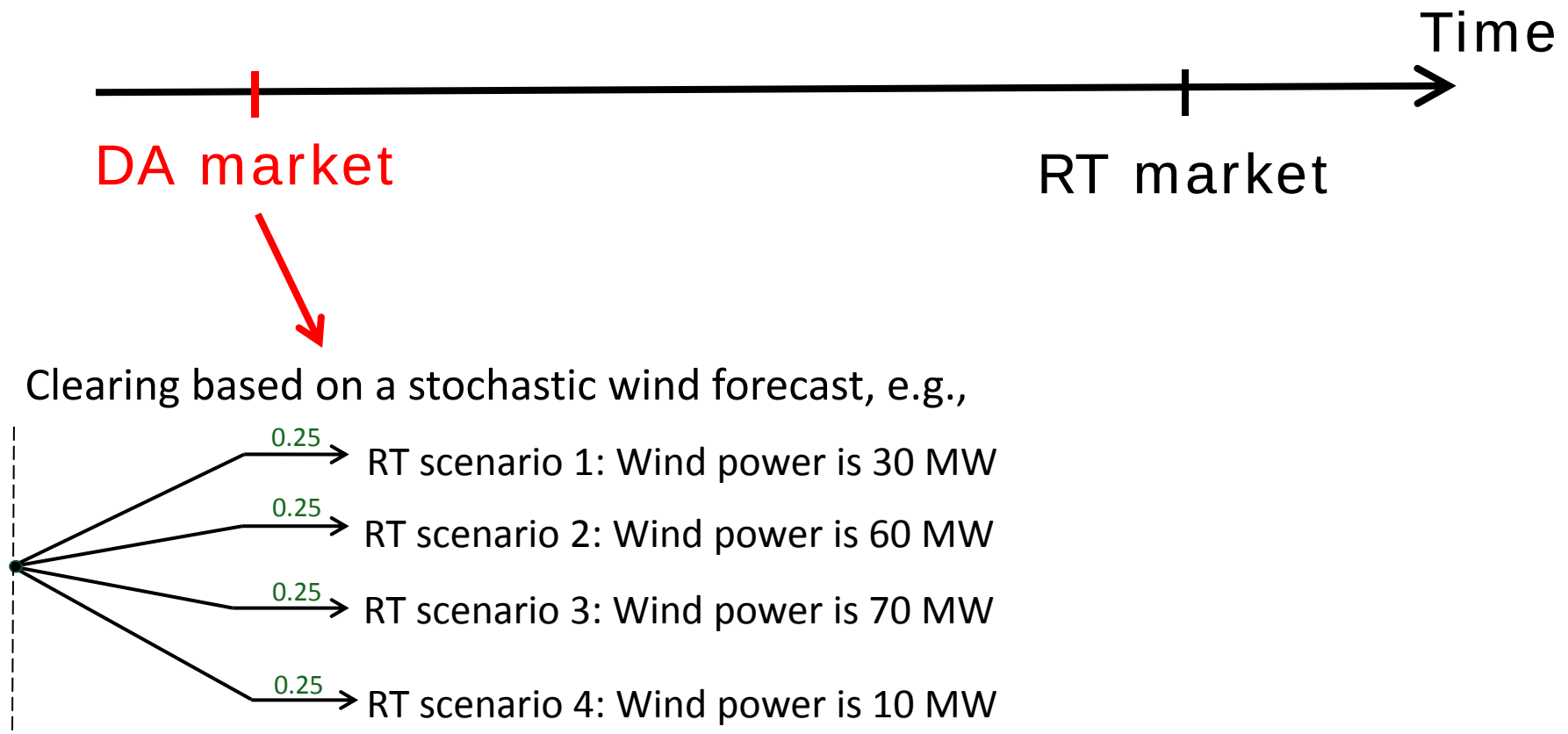
Deterministic vs stochastic market clearing

In a **deterministic** market mechanism:



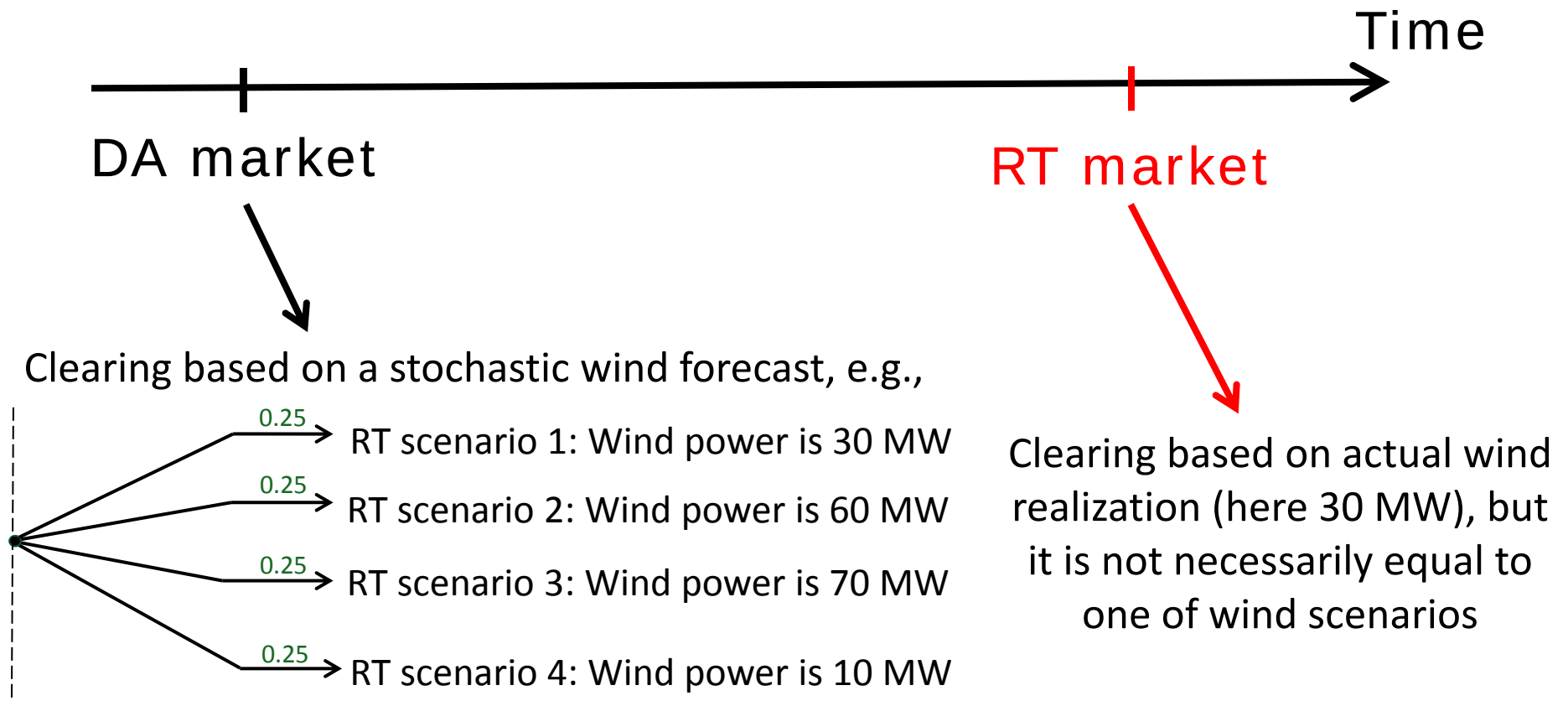
Deterministic vs stochastic market clearing

In a stochastic market mechanism:



Deterministic vs stochastic market clearing

In a **stochastic** market mechanism:



Stochastic market clearing: optimization



Stochastic market clearing: optimization

Minimize (total operational cost¹ of the system in DA) +
(total **expected** operational cost of the system in RT)

subject to:

- Production limits (in DA and RT)
- Transmission network limits (in DA and RT)
- Load shedding limits (in RT)
- Nodal power balances (in DA and RT)

¹ We assume loads are inelastic to price, otherwise the social welfare should be maximized.

Stochastic market clearing: optimization

Minimize (total operational cost¹ of the system in DA) +
(total **expected** operational cost of the system in RT)

subject to:

- Production limits (in DA and RT)
- Transmission network limits (in DA and RT)
- Load shedding limits (in RT)
- Nodal power balances (in DA and RT)

¹ We assume loads are inelastic to price, otherwise the social welfare should be maximized.

Relevant references:

- [1] G. Pritchard, G. Zakeri, and A. Philpott, "A single-settlement, energy-only electric power market or unpredictable and intermittent participants," *Operations Research*, vol. 58, no. 4, pp. 1210-1219, 2010.
- [2] J. M. Morales, A. J. Conejo, K. Liu, and J. Zhong, "Pricing electricity in pools with wind producers," *IEEE Transactions on Power Systems*, vol. 27, no.3, pp. 1366-1376, 2012.

Stochastic market clearing: optimization

Recall this example:



- Capacity: 100 MW
- Production cost: \$10/MWh
- Inflexible in real time (RT)



- Capacity: 30 MW
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- Fully flexible in RT



- Capacity: 70 MW
- Production cost: \$0/MWh
- Uncertain production in DA
- Realization in RT: 30 MW



- 120 MW (inelastic and deterministic)
- Curtailment cost: \$80/MWh

Stochastic market clearing: optimization

$$\begin{aligned} & \text{Minimize} \\ & p^{G1,DA}, p^{G2,DA}, p^{WP,DA}, p_{\omega_1}^{G2,RT}, p_{\omega_2}^{G2,RT}, p_{\omega_3}^{G2,RT}, p_{\omega_4}^{G2,RT}, p_{\omega_1}^{shed}, p_{\omega_2}^{shed}, p_{\omega_3}^{shed}, p_{\omega_4}^{shed}, p_{\omega_1}^{spill}, p_{\omega_2}^{spill}, p_{\omega_3}^{spill}, p_{\omega_4}^{spill} \\ & [10p^{G1,DA} + 20p^{G2,DA}] + 0.25 \left[20p_{\omega_1}^{G2,RT} + 80p_{\omega_1}^{shed} + 20p_{\omega_2}^{G2,RT} + 80p_{\omega_2}^{shed} + \right. \\ & \quad \left. 20p_{\omega_3}^{G2,RT} + 80p_{\omega_3}^{shed} + 20p_{\omega_4}^{G2,RT} + 80p_{\omega_4}^{shed} \right] \end{aligned}$$

subject to:

$$0 \leq p^{G1,DA} \leq 100$$

$$0 \leq p^{G2,DA} \leq 30$$

$$0 \leq p^{WP,DA} \leq 70$$

$$p^{G1,DA} + p^{G2,DA} + p^{WP,DA} = 120 \quad : \lambda^{DA}$$

$$0 \leq (p^{G2,DA} + p_{\omega_1}^{G2,RT}) \leq 30$$

$$-30 \leq p_{\omega_1}^{G2,RT} \leq 30$$

$$0 \leq p_{\omega_1}^{spill} \leq 30$$

$$0 \leq p_{\omega_1}^{shed} \leq 120$$

$$p_{\omega_1}^{G2,RT} + (30 - p^{WP,DA} - p_{\omega_1}^{spill}) + p_{\omega_1}^{shed} = 0 \quad : \lambda_{\omega_1}^{RT}$$

Stochastic market clearing: optimization

DA variables: schedules of G1, G2 and WP

$$\begin{aligned} & \text{Minimize} \\ & p^{G1,DA}, p^{G2,DA}, p^{WP,DA}, p_{\omega_1}^{G2,RT}, p_{\omega_2}^{G2,RT}, p_{\omega_3}^{G2,RT}, p_{\omega_4}^{G2,RT}, p_{\omega_1}^{shed}, p_{\omega_2}^{shed}, p_{\omega_3}^{shed}, p_{\omega_4}^{shed}, p_{\omega_1}^{spill}, p_{\omega_2}^{spill}, p_{\omega_3}^{spill}, p_{\omega_4}^{spill} \\ & [10p^{G1,DA} + 20p^{G2,DA}] + 0.25 \left[20p_{\omega_1}^{G2,RT} + 80p_{\omega_1}^{shed} + 20p_{\omega_2}^{G2,RT} + 80p_{\omega_2}^{shed} + \right. \\ & \quad \left. 20p_{\omega_3}^{G2,RT} + 80p_{\omega_3}^{shed} + 20p_{\omega_4}^{G2,RT} + 80p_{\omega_4}^{shed} \right] \end{aligned}$$

subject to:

$$0 \leq p^{G1,DA} \leq 100$$

$$0 \leq p^{G2,DA} \leq 30$$

$$0 \leq p^{WP,DA} \leq 70$$

$$p^{G1,DA} + p^{G2,DA} + p^{WP,DA} = 120 \quad : \lambda^{DA}$$

$$0 \leq (p^{G2,DA} + p_{\omega_1}^{G2,RT}) \leq 30$$

$$-30 \leq p_{\omega_1}^{G2,RT} \leq 30$$

$$0 \leq p_{\omega_1}^{spill} \leq 30$$

$$0 \leq p_{\omega_1}^{shed} \leq 120$$

$$p_{\omega_1}^{G2,RT} + (30 - p^{WP,DA} - p_{\omega_1}^{spill}) + p_{\omega_1}^{shed} = 0 \quad : \lambda_{\omega_1}^{RT}$$

Stochastic market clearing: optimization

RT variables: flexible unit G2's adjustment under scenarios w1 to w4

$$\begin{aligned} & \text{Minimize} \\ & p^{G1,DA}, p^{G2,DA}, p^{WP,DA}, p_{\omega_1}^{G2,RT}, p_{\omega_2}^{G2,RT}, p_{\omega_3}^{G2,RT}, p_{\omega_4}^{G2,RT}, p_{\omega_1}^{shed}, p_{\omega_2}^{shed}, p_{\omega_3}^{shed}, p_{\omega_4}^{shed}, p_{\omega_1}^{spill}, p_{\omega_2}^{spill}, p_{\omega_3}^{spill}, p_{\omega_4}^{spill} \\ & [10p^{G1,DA} + 20p^{G2,DA}] + 0.25 \left[20p_{\omega_1}^{G2,RT} + 80p_{\omega_1}^{shed} + 20p_{\omega_2}^{G2,RT} + 80p_{\omega_2}^{shed} + \right. \\ & \quad \left. 20p_{\omega_3}^{G2,RT} + 80p_{\omega_3}^{shed} + 20p_{\omega_4}^{G2,RT} + 80p_{\omega_4}^{shed} \right] \end{aligned}$$

subject to:

$$0 \leq p^{G1,DA} \leq 100$$

$$0 \leq p^{G2,DA} \leq 30$$

$$0 \leq p^{WP,DA} \leq 70$$

$$p^{G1,DA} + p^{G2,DA} + p^{WP,DA} = 120 \quad : \lambda^{DA}$$

$$0 \leq (p^{G2,DA} + p_{\omega_1}^{G2,RT}) \leq 30$$

$$-30 \leq p_{\omega_1}^{G2,RT} \leq 30$$

$$0 \leq p_{\omega_1}^{spill} \leq 30$$

$$0 \leq p_{\omega_1}^{shed} \leq 120$$

$$p_{\omega_1}^{G2,RT} + (30 - p^{WP,DA} - p_{\omega_1}^{spill}) + p_{\omega_1}^{shed} = 0 \quad : \lambda_{\omega_1}^{RT}$$

Stochastic market clearing: optimization

RT variables: load curtailment under scenarios w1 to w4

$$\begin{aligned} & \text{Minimize} \\ & p^{G1,DA}, p^{G2,DA}, p^{WP,DA}, p_{\omega_1}^{G2,RT}, p_{\omega_2}^{G2,RT}, p_{\omega_3}^{G2,RT}, p_{\omega_4}^{G2,RT}, p_{\omega_1}^{shed}, p_{\omega_2}^{shed}, p_{\omega_3}^{shed}, p_{\omega_4}^{shed}, p_{\omega_1}^{spill}, p_{\omega_2}^{spill}, p_{\omega_3}^{spill}, p_{\omega_4}^{spill} \\ & [10p^{G1,DA} + 20p^{G2,DA}] + 0.25 \left[20p_{\omega_1}^{G2,RT} + 80p_{\omega_1}^{shed} + 20p_{\omega_2}^{G2,RT} + 80p_{\omega_2}^{shed} + \right. \\ & \quad \left. 20p_{\omega_3}^{G2,RT} + 80p_{\omega_3}^{shed} + 20p_{\omega_4}^{G2,RT} + 80p_{\omega_4}^{shed} \right] \end{aligned}$$

subject to:

$$0 \leq p^{G1,DA} \leq 100$$

$$0 \leq p^{G2,DA} \leq 30$$

$$0 \leq p^{WP,DA} \leq 70$$

$$p^{G1,DA} + p^{G2,DA} + p^{WP,DA} = 120 \quad : \lambda^{DA}$$

$$0 \leq (p^{G2,DA} + p_{\omega_1}^{G2,RT}) \leq 30$$

$$-30 \leq p_{\omega_1}^{G2,RT} \leq 30$$

$$0 \leq p_{\omega_1}^{spill} \leq 30$$

$$0 \leq p_{\omega_1}^{shed} \leq 120$$

$$p_{\omega_1}^{G2,RT} + (30 - p^{WP,DA} - p_{\omega_1}^{spill}) + p_{\omega_1}^{shed} = 0 \quad : \lambda_{\omega_1}^{RT}$$

Stochastic market clearing: optimization

RT variables: wind power spillage under scenarios w1 to w4

$$\begin{aligned} & \text{Minimize} \\ & p^{G1,DA}, p^{G2,DA}, p^{WP,DA}, p_{\omega_1}^{G2,RT}, p_{\omega_2}^{G2,RT}, p_{\omega_3}^{G2,RT}, p_{\omega_4}^{G2,RT}, p_{\omega_1}^{shed}, p_{\omega_2}^{shed}, p_{\omega_3}^{shed}, p_{\omega_4}^{shed}, \boxed{p_{\omega_1}^{spill}, p_{\omega_2}^{spill}, p_{\omega_3}^{spill}, p_{\omega_4}^{spill}} \\ & [10p^{G1,DA} + 20p^{G2,DA}] + 0.25 \left[20p_{\omega_1}^{G2,RT} + 80p_{\omega_1}^{shed} + 20p_{\omega_2}^{G2,RT} + 80p_{\omega_2}^{shed} + \right. \\ & \quad \left. 20p_{\omega_3}^{G2,RT} + 80p_{\omega_3}^{shed} + 20p_{\omega_4}^{G2,RT} + 80p_{\omega_4}^{shed} \right] \end{aligned}$$

subject to:

$$0 \leq p^{G1,DA} \leq 100$$

$$0 \leq p^{G2,DA} \leq 30$$

$$0 \leq p^{WP,DA} \leq 70$$

$$p^{G1,DA} + p^{G2,DA} + p^{WP,DA} = 120 \quad : \lambda^{DA}$$

$$0 \leq (p^{G2,DA} + p_{\omega_1}^{G2,RT}) \leq 30$$

$$-30 \leq p_{\omega_1}^{G2,RT} \leq 30$$

$$0 \leq p_{\omega_1}^{spill} \leq 30$$

$$0 \leq p_{\omega_1}^{shed} \leq 120$$

$$p_{\omega_1}^{G2,RT} + (30 - p^{WP,DA} - p_{\omega_1}^{spill}) + p_{\omega_1}^{shed} = 0 \quad : \lambda_{\omega_1}^{RT}$$

Stochastic market clearing: optimization

Minimize

$$p^{G1,DA}, p^{G2,DA}, p^{WP,DA}, p_{\omega_1}^{G2,RT}, p_{\omega_2}^{G2,RT}, p_{\omega_3}^{G2,RT}, p_{\omega_4}^{G2,RT}, p_{\omega_1}^{shed}, p_{\omega_2}^{shed}, p_{\omega_3}^{shed}, p_{\omega_4}^{shed}, p_{\omega_1}^{spill}, p_{\omega_2}^{spill}, p_{\omega_3}^{spill}, p_{\omega_4}^{spill}$$

$$\boxed{[10p^{G1,DA} + 20p^{G2,DA}]} + 0.25 \left[20p_{\omega_1}^{G2,RT} + 80p_{\omega_1}^{shed} + 20p_{\omega_2}^{G2,RT} + 80p_{\omega_2}^{shed} + 20p_{\omega_3}^{G2,RT} + 80p_{\omega_3}^{shed} + 20p_{\omega_4}^{G2,RT} + 80p_{\omega_4}^{shed} \right]$$

subject to:

$$0 \leq p^{G1,DA} \leq 100$$

$$0 \leq p^{G2,DA} \leq 30$$

$$0 \leq p^{WP,DA} \leq 70$$

$$p^{G1,DA} + p^{G2,DA} + p^{WP,DA} = 120 \quad : \lambda^{DA}$$

$$0 \leq (p^{G2,DA} + p_{\omega_1}^{G2,RT}) \leq 30$$

$$-30 \leq p_{\omega_1}^{G2,RT} \leq 30$$

$$0 \leq p_{\omega_1}^{spill} \leq 30$$

$$0 \leq p_{\omega_1}^{shed} \leq 120$$

$$p_{\omega_1}^{G2,RT} + (30 - p^{WP,DA} - p_{\omega_1}^{spill}) + p_{\omega_1}^{shed} = 0 \quad : \lambda_{\omega_1}^{RT}$$

Total operational cost of the system in DA

Total expected operational cost of the system in RT

Stochastic market clearing: optimization

$$\begin{aligned} & \text{Minimize} \\ & p^{G1,DA}, p^{G2,DA}, p^{WP,DA}, p_{\omega_1}^{G2,RT}, p_{\omega_2}^{G2,RT}, p_{\omega_3}^{G2,RT}, p_{\omega_4}^{G2,RT}, p_{\omega_1}^{shed}, p_{\omega_2}^{shed}, p_{\omega_3}^{shed}, p_{\omega_4}^{shed}, p_{\omega_1}^{spill}, p_{\omega_2}^{spill}, p_{\omega_3}^{spill}, p_{\omega_4}^{spill} \\ & [10p^{G1,DA} + 20p^{G2,DA}] + 0.25 \left[20p_{\omega_1}^{G2,RT} + 80p_{\omega_1}^{shed} + 20p_{\omega_2}^{G2,RT} + 80p_{\omega_2}^{shed} + \right. \\ & \quad \left. 20p_{\omega_3}^{G2,RT} + 80p_{\omega_3}^{shed} + 20p_{\omega_4}^{G2,RT} + 80p_{\omega_4}^{shed} \right] \end{aligned}$$

subject to:

$$0 \leq p^{G1,DA} \leq 100$$

$$0 \leq p^{G2,DA} \leq 30$$

$$0 \leq p^{WP,DA} \leq 70$$

$$p^{G1,DA} + p^{G2,DA} + p^{WP,DA} = 120 \quad : \lambda^{DA}$$

DA constraints

$$0 \leq (p^{G2,DA} + p_{\omega_1}^{G2,RT}) \leq 30$$

$$-30 \leq p_{\omega_1}^{G2,RT} \leq 30$$

$$0 \leq p_{\omega_1}^{spill} \leq 30$$

$$0 \leq p_{\omega_1}^{shed} \leq 120$$

$$p_{\omega_1}^{G2,RT} + (30 - p^{WP,DA} - p_{\omega_1}^{spill}) + p_{\omega_1}^{shed} = 0 \quad : \lambda_{\omega_1}^{RT}$$

Stochastic market clearing: optimization

$$\begin{aligned} & \text{Minimize} \\ & p^{G1,DA}, p^{G2,DA}, p^{WP,DA}, p_{\omega_1}^{G2,RT}, p_{\omega_2}^{G2,RT}, p_{\omega_3}^{G2,RT}, p_{\omega_4}^{G2,RT}, p_{\omega_1}^{shed}, p_{\omega_2}^{shed}, p_{\omega_3}^{shed}, p_{\omega_4}^{shed}, p_{\omega_1}^{spill}, p_{\omega_2}^{spill}, p_{\omega_3}^{spill}, p_{\omega_4}^{spill} \\ & [10p^{G1,DA} + 20p^{G2,DA}] + 0.25 \left[20p_{\omega_1}^{G2,RT} + 80p_{\omega_1}^{shed} + 20p_{\omega_2}^{G2,RT} + 80p_{\omega_2}^{shed} + \right. \\ & \quad \left. 20p_{\omega_3}^{G2,RT} + 80p_{\omega_3}^{shed} + 20p_{\omega_4}^{G2,RT} + 80p_{\omega_4}^{shed} \right] \end{aligned}$$

subject to:

$$0 \leq p^{G1,DA} \leq 100$$

$$0 \leq p^{G2,DA} \leq 30$$

$$0 \leq p^{WP,DA} \leq 70$$

Installed
capacity of
wind farm

DA constraints

$$p^{G1,DA} + p^{G2,DA} + p^{WP,DA} = 120 \quad : \lambda^{DA}$$

$$0 \leq (p^{G2,DA} + p_{\omega_1}^{G2,RT}) \leq 30$$

$$-30 \leq p_{\omega_1}^{G2,RT} \leq 30$$

$$0 \leq p_{\omega_1}^{spill} \leq 30$$

$$0 \leq p_{\omega_1}^{shed} \leq 120$$

$$p_{\omega_1}^{G2,RT} + (30 - p^{WP,DA} - p_{\omega_1}^{spill}) + p_{\omega_1}^{shed} = 0 \quad : \lambda_{\omega_1}^{RT}$$

Stochastic market clearing: optimization

$$\begin{aligned} & \text{Minimize} \\ & p^{G1,DA}, p^{G2,DA}, p^{WP,DA}, p_{\omega_1}^{G2,RT}, p_{\omega_2}^{G2,RT}, p_{\omega_3}^{G2,RT}, p_{\omega_4}^{G2,RT}, p_{\omega_1}^{shed}, p_{\omega_2}^{shed}, p_{\omega_3}^{shed}, p_{\omega_4}^{shed}, p_{\omega_1}^{spill}, p_{\omega_2}^{spill}, p_{\omega_3}^{spill}, p_{\omega_4}^{spill} \\ & [10p^{G1,DA} + 20p^{G2,DA}] + 0.25 \left[20p_{\omega_1}^{G2,RT} + 80p_{\omega_1}^{shed} + 20p_{\omega_2}^{G2,RT} + 80p_{\omega_2}^{shed} + \right. \\ & \quad \left. 20p_{\omega_3}^{G2,RT} + 80p_{\omega_3}^{shed} + 20p_{\omega_4}^{G2,RT} + 80p_{\omega_4}^{shed} \right] \end{aligned}$$

subject to:

$$0 \leq p^{G1,DA} \leq 100$$

$$0 \leq p^{G2,DA} \leq 30$$

$$0 \leq p^{WP,DA} \leq 70$$

Installed
capacity of
wind farm

DA constraints

$$p^{G1,DA} + p^{G2,DA} + p^{WP,DA} = 120 \quad : \lambda^{DA}$$

$$0 \leq (p^{G2,DA} + p_{\omega_1}^{G2,RT}) \leq 30$$

$$-30 \leq p_{\omega_1}^{G2,RT} \leq 30$$

$$0 \leq p_{\omega_1}^{spill} \leq 30$$

$$0 \leq p_{\omega_1}^{shed} \leq 120$$

$$p_{\omega_1}^{G2,RT} + (30 - p^{WP,DA} - p_{\omega_1}^{spill}) + p_{\omega_1}^{shed} = 0 \quad : \lambda_{\omega_1}^{RT}$$

Discussion:

Any other potential alternative for the upper bound of the wind power schedule in DA?

Stochastic market clearing: optimization

$$\begin{aligned} & \text{Minimize} \\ & p^{G1,DA}, p^{G2,DA}, p^{WP,DA}, p_{\omega_1}^{G2,RT}, p_{\omega_2}^{G2,RT}, p_{\omega_3}^{G2,RT}, p_{\omega_4}^{G2,RT}, p_{\omega_1}^{shed}, p_{\omega_2}^{shed}, p_{\omega_3}^{shed}, p_{\omega_4}^{shed}, p_{\omega_1}^{spill}, p_{\omega_2}^{spill}, p_{\omega_3}^{spill}, p_{\omega_4}^{spill} \\ & [10p^{G1,DA} + 20p^{G2,DA}] + 0.25 \left[20p_{\omega_1}^{G2,RT} + 80p_{\omega_1}^{shed} + 20p_{\omega_2}^{G2,RT} + 80p_{\omega_2}^{shed} + \right. \\ & \quad \left. 20p_{\omega_3}^{G2,RT} + 80p_{\omega_3}^{shed} + 20p_{\omega_4}^{G2,RT} + 80p_{\omega_4}^{shed} \right] \end{aligned}$$

subject to:

$$0 \leq p^{G1,DA} \leq 100$$

$$0 \leq p^{G2,DA} \leq 30$$

$$0 \leq p^{WP,DA} \leq 70$$

$$p^{G1,DA} + p^{G2,DA} + p^{WP,DA} = 120 \quad : \lambda^{DA}$$

$$0 \leq (p^{G2,DA} + p_{\omega_1}^{G2,RT}) \leq 30$$

$$-30 \leq p_{\omega_1}^{G2,RT} \leq 30$$

$$0 \leq p_{\omega_1}^{spill} \leq 30$$

$$0 \leq p_{\omega_1}^{shed} \leq 120$$

$$p_{\omega_1}^{G2,RT} + (30 - p^{WP,DA} - p_{\omega_1}^{spill}) + p_{\omega_1}^{shed} = 0 \quad : \lambda_{\omega_1}^{RT}$$

RT constraints under scenario w1
(similar constraints are needed for scenarios w2 to w4)

Stochastic market clearing: optimization

$$\begin{aligned} & \text{Minimize} \\ & p^{G1,DA}, p^{G2,DA}, p^{WP,DA}, p_{\omega_1}^{G2,RT}, p_{\omega_2}^{G2,RT}, p_{\omega_3}^{G2,RT}, p_{\omega_4}^{G2,RT}, p_{\omega_1}^{shed}, p_{\omega_2}^{shed}, p_{\omega_3}^{shed}, p_{\omega_4}^{shed}, p_{\omega_1}^{spill}, p_{\omega_2}^{spill}, p_{\omega_3}^{spill}, p_{\omega_4}^{spill} \\ & [10p^{G1,DA} + 20p^{G2,DA}] + 0.25 \left[20p_{\omega_1}^{G2,RT} + 80p_{\omega_1}^{shed} + 20p_{\omega_2}^{G2,RT} + 80p_{\omega_2}^{shed} + \right. \\ & \quad \left. 20p_{\omega_3}^{G2,RT} + 80p_{\omega_3}^{shed} + 20p_{\omega_4}^{G2,RT} + 80p_{\omega_4}^{shed} \right] \end{aligned}$$

subject to:

$$0 \leq p^{G1,DA} \leq 100$$

$$0 \leq p^{G2,DA} \leq 30$$

$$0 \leq p^{WP,DA} \leq 70$$

$$p^{G1,DA} + p^{G2,DA} + p^{WP,DA} = 120 \quad : \lambda^{DA}$$

$$0 \leq (p^{G2,DA} + p_{\omega_1}^{G2,RT}) \leq 30$$

$$-30 \leq p_{\omega_1}^{G2,RT} \leq 30$$

$$0 \leq p_{\omega_1}^{spill} \leq 30$$

$$0 \leq p_{\omega_1}^{shed} \leq 120$$

$$p_{\omega_1}^{G2,RT} + (30 - p^{WP,DA} - p_{\omega_1}^{spill}) + p_{\omega_1}^{shed} = 0 \quad : \lambda_{\omega_1}^{RT}$$

RT constraints under scenario w1
(similar constraints are needed for scenarios w2 to w4)

Discussion:
Please interpret each RT constraint!

Stochastic market clearing: optimization

$$\begin{aligned} & \text{Minimize} \\ & p^{G1,DA}, p^{G2,DA}, p^{WP,DA}, p_{\omega_1}^{G2,RT}, p_{\omega_2}^{G2,RT}, p_{\omega_3}^{G2,RT}, p_{\omega_4}^{G2,RT}, p_{\omega_1}^{shed}, p_{\omega_2}^{shed}, p_{\omega_3}^{shed}, p_{\omega_4}^{shed}, p_{\omega_1}^{spill}, p_{\omega_2}^{spill}, p_{\omega_3}^{spill}, p_{\omega_4}^{spill} \\ & [10p^{G1,DA} + 20p^{G2,DA}] + 0.25 \left[20p_{\omega_1}^{G2,RT} + 80p_{\omega_1}^{shed} + 20p_{\omega_2}^{G2,RT} + 80p_{\omega_2}^{shed} + \right. \\ & \quad \left. 20p_{\omega_3}^{G2,RT} + 80p_{\omega_3}^{shed} + 20p_{\omega_4}^{G2,RT} + 80p_{\omega_4}^{shed} \right] \end{aligned}$$

subject to:

$$0 \leq p^{G1,DA} \leq 100$$

$$0 \leq p^{G2,DA} \leq 30$$

$$0 \leq p^{WP,DA} \leq 70$$

$$p^{G1,DA} + p^{G2,DA} + p^{WP,DA} = 120 \quad : \lambda^{DA}$$

$$0 \leq (p^{G2,DA} + p_{\omega_1}^{G2,RT}) \leq 30$$

$$-30 \leq p_{\omega_1}^{G2,RT} \leq 30$$

$$0 \leq p_{\omega_1}^{spill} \leq 30 \quad \text{Wind power realization}$$

$$0 \leq p_{\omega_1}^{shed} \leq 120 \quad \text{under scenario w1}$$

$$p_{\omega_1}^{G2,RT} + (30 - p^{WP,DA} - p_{\omega_1}^{spill}) + p_{\omega_1}^{shed} = 0 \quad : \lambda_{\omega_1}^{RT}$$

Stochastic market clearing: optimization

$$\begin{aligned} & \text{Minimize} \\ & p^{G1,DA}, p^{G2,DA}, p^{WP,DA}, p_{\omega_1}^{G2,RT}, p_{\omega_2}^{G2,RT}, p_{\omega_3}^{G2,RT}, p_{\omega_4}^{G2,RT}, p_{\omega_1}^{shed}, p_{\omega_2}^{shed}, p_{\omega_3}^{shed}, p_{\omega_4}^{shed}, p_{\omega_1}^{spill}, p_{\omega_2}^{spill}, p_{\omega_3}^{spill}, p_{\omega_4}^{spill} \\ & [10p^{G1,DA} + 20p^{G2,DA}] + 0.25 \left[20p_{\omega_1}^{G2,RT} + 80p_{\omega_1}^{shed} + 20p_{\omega_2}^{G2,RT} + 80p_{\omega_2}^{shed} + \right. \\ & \quad \left. 20p_{\omega_3}^{G2,RT} + 80p_{\omega_3}^{shed} + 20p_{\omega_4}^{G2,RT} + 80p_{\omega_4}^{shed} \right] \end{aligned}$$

subject to:

$$0 \leq p^{G1,DA} \leq 100$$

$$0 \leq p^{G2,DA} \leq 30$$

$$0 \leq p^{WP,DA} \leq 70$$

$$p^{G1,DA} + p^{G2,DA} + p^{WP,DA} = 120 \quad : \lambda^{DA}$$

$$0 \leq (p^{G2,DA} + p_{\omega_1}^{G2,RT}) \leq 30$$

$$-30 \leq p_{\omega_1}^{G2,RT} \leq 30$$

Constraints linking DA
and RT variables

$$0 \leq p_{\omega_1}^{spill} \leq 30$$

$$0 \leq p_{\omega_1}^{shed} \leq 120$$

$$p_{\omega_1}^{G2,RT} + (30 - p^{WP,DA} - p_{\omega_1}^{spill}) + p_{\omega_1}^{shed} = 0 \quad : \lambda_{\omega_1}^{RT}$$

Stochastic market clearing: optimization

$$\begin{aligned} & \text{Minimize} \\ & p^{G1,DA}, p^{G2,DA}, p^{WP,DA}, p_{\omega_1}^{G2,RT}, p_{\omega_2}^{G2,RT}, p_{\omega_3}^{G2,RT}, p_{\omega_4}^{G2,RT}, p_{\omega_1}^{shed}, p_{\omega_2}^{shed}, p_{\omega_3}^{shed}, p_{\omega_4}^{shed}, p_{\omega_1}^{spill}, p_{\omega_2}^{spill}, p_{\omega_3}^{spill}, p_{\omega_4}^{spill} \\ & [10p^{G1,DA} + 20p^{G2,DA}] + 0.25 \left[20p_{\omega_1}^{G2,RT} + 80p_{\omega_1}^{shed} + 20p_{\omega_2}^{G2,RT} + 80p_{\omega_2}^{shed} + \right. \\ & \quad \left. 20p_{\omega_3}^{G2,RT} + 80p_{\omega_3}^{shed} + 20p_{\omega_4}^{G2,RT} + 80p_{\omega_4}^{shed} \right] \end{aligned}$$

subject to:

$$0 \leq p^{G1,DA} \leq 100$$

$$0 \leq p^{G2,DA} \leq 30$$

$$0 \leq p^{WP,DA} \leq 70$$

$$p^{G1,DA} + p^{G2,DA} + p^{WP,DA} = 120 \quad \boxed{\lambda^{DA}} \text{ DA market price}$$

$$0 \leq (p^{G2,DA} + p_{\omega_1}^{G2,RT}) \leq 30$$

$$-30 \leq p_{\omega_1}^{G2,RT} \leq 30$$

$$0 \leq p_{\omega_1}^{spill} \leq 30$$

$$0 \leq p_{\omega_1}^{shed} \leq 120$$

$$p_{\omega_1}^{G2,RT} + (30 - p^{WP,DA} - p_{\omega_1}^{spill}) + p_{\omega_1}^{shed} = 0$$

$$\boxed{\lambda_{\omega_1}^{RT}}$$

Probability-weighted

RT market price
under scenario w1

Stochastic market clearing: optimization

	DA	RT (scenario 1) Wind power realization: 30 MW	RT (scenario 2) Wind power realization: 60 MW	RT (scenario 3) Wind power realization: 70 MW	RT (scenario 4) Wind power realization: 10 MW
G1 [MW]	80				
G2 [MW]	30				
WP [MW]	10				
Load [MW]	120				
Corresponding operational cost [\$]	1400				

Stochastic market clearing: optimization

	DA	RT (scenario 1) Wind power realization: 30 MW	RT (scenario 2) Wind power realization: 60 MW	RT (scenario 3) Wind power realization: 70 MW	RT (scenario 4) Wind power realization: 10 MW
G1 [MW]	80	unchanged	unchanged	unchanged	unchanged
G2 [MW]	30	-20 (down regulation)	-30 (down regulation)	-30 (down regulation)	unchanged
WP [MW]	10	+20	+30 (20 MW spilled)	+30 (30 MW spilled)	unchanged
Load [MW]	120	unchanged (not curtailed)	unchanged (not curtailed)	unchanged (not curtailed)	unchanged (not curtailed)
Corresponding operational cost [\$]	1400	-400	-600	-600	0

Stochastic market clearing: optimization

	DA	RT (scenario 1) Wind power realization: 30 MW	RT (scenario 2) Wind power realization: 60 MW	RT (scenario 3) Wind power realization: 70 MW	RT (scenario 4) Wind power realization: 10 MW
G1 [MW]	80	unchanged	unchanged	unchanged	unchanged
G2 [MW]	30	-20 (down regulation)	-30 (down regulation)	-30 (down regulation)	unchanged
WP [MW]	10	+20	+30 (20 MW spilled)	+30 (30 MW spilled)	unchanged
Load [MW]	120	unchanged (not curtailed)	unchanged (not curtailed)	unchanged (not curtailed)	unchanged (not curtailed)
Corresponding operational cost [\$]	1400	-400	-600	-600	0

$$\begin{aligned}
 &\text{Total } \underline{\text{expected}} \text{ operational cost of the system} \\
 &= 1400 + 0.25 * [-400 - 600 - 600 + 0] \\
 &= \$1000
 \end{aligned}$$

GAMS code

```

sets
i   conventional generators /i1*i2/
d   inelastic loads /d1/
n   buses /n1*n2/
s   scenarios /s1*s4/
k   wind power generators /k1/
slack(n)   /n1/
Mapi(i,n)  /i1.n1 , i2.n2/
Mapd(d,n)  /d1.n1/
Mapnm(n,n) /n1.n2 , n2.n1/
Mapk(k,n)  /k1.n2/
alias (n,m);

parameters
W_max(k)   Wind power installed capacity;

parameters
phi(s) probability of scenarios /
s1  0.25
s2  0.25
s3  0.25
s4  0.25/

L(d) Load level /d1 120/
V(d) value of lost load /d1 80/;

```

GAMS code

```

TABLE GDATA(i,*) generators' input data
      Pmax      C      Rmax
i1      100      10      0
i2      30      20      30;

Table W(k,s) Wind realizatio under different scenarios
      s1      s2      s3      s4
k1      30      60      70      10;

Table Fmax(n,n) Transmission lines capacity
      n1      n2
n1      0      100
n2      100      0;

Table B(n,n) Transmission lines susceptance
      n1      n2
n1      0      500
n2      500      0;

W_max(k)=70;

variables
cost          Total expected system cost (DA + RT)
theta_DA(n)   Voltage angles in DA
f_DA(n,m)     Power flows in DA
f_RT(n,m,s)   Power flows in RT
theta_RT(n,s) Voltage angles in RT
time          CPU time
r(i,s)        Power adjustment of generator i in RT under scenario s;

positive variables
L_shed(d,s)   Curtailed load
P(i)          DA dispatch of generators
P_W(k)        Wind dispatch in DA
P_spill(k,s)  Wind spillage;

```

GAMS code

equations

```
costfn,node_DA,Pmax,kmax,flow_DA,flow_max_DA,slack_DA,node_RT,RUmax,RDmax,
generation_min,generation_max,flow_RT,flow_max_RT,shedding,spillage,
slack_RT;
```

```
costfn..cost=e=sum(s,phi(s)*{
    [sum(i,GDATA(i,'C')*r(i,s))]
    +[sum(d,V(d)*L_shed(d,s))]})+[sum(i,GDATA(i,'C')*P(i))];
```

* DA constraints

```
node_DA(n)..sum(i$Mapi(i,n),P(i))+sum(k$Mapk(k,n),P_W(k))
    -sum(d$Mapd(d,n),L(d))-sum(m$Mapnm(n,m),f_DA(n,m))=e=0;
```

```
Pmax(i)..P(i)=l=GDATA(i,'Pmax');
kmax(k)..P_W(k)=l=W_max(k);
flow_DA(n,m)$Mapnm(n,m)..f_DA(n,m)=e=B(n,m)*(theta_DA(n)-theta_DA(m));
flow_max_DA(n,m)$Mapnm(n,m)..f_DA(n,m)=l=Fmax(n,m);
slack_DA..theta_DA('n1')=e=0;
```

GAMS code

```
* RT constraints
node_RT(n,s)..sum(i$Mapi(i,n),r(i,s))
               +sum(k$Mapk(k,n),W(k,s)-P_W(k)-P_spill(k,s))
               +sum(d$Mapd(d,n),L_shed(d,s))
               -sum(m$Mapnm(n,m),f_RT(n,m,s)-f_DA(n,m))=e=0;

RUmax(i,s)..r(i,s)=l=+GDATA(i,'Rmax');
RDmax(i,s)..r(i,s)=g=-GDATA(i,'Rmax');
generation_min(i,s)..[P(i)+r(i,s)]=g=0;
generation_max(i,s)..[P(i)+r(i,s)]=l=GDATA(i,'Pmax');
flow_RT(n,m,s)$Mapnm(n,m)..f_RT(n,m,s)=e=B(n,m)*(theta_RT(n,s)-theta_RT(m,s));
flow_max_RT(n,m,s)$Mapnm(n,m)..f_RT(n,m,s)=l=Fmax(n,m);
shedding(d,s)..L_shed(d,s)=l=L(d);
spillage(k,s)..P_spill(k,s)=l=W(k,s);
slack_RT(s)..theta_RT('n1',s)=e=0;

model Stochastic_clearing / all / ;
solve Stochastic_clearing using lp minimizing cost;
time.l = Stochastic_clearing.resusd;

parameters
LambdaN_DA(n)
LambdaN_RT(n,s);
LambdaN_DA(n)=node_DA.m(n);
LambdaN_RT(n,s)=node_RT.m(n,s)/phi(s);

display
cost.l, LambdaN_DA, LambdaN_RT, P_W.l, P.l, r.l, p_spill.l, L_shed.l, time.l;
```

Exercise 1

In the previous numerical example, derive and interpret market-clearing prices in DA and in RT under each scenario.

Exercise 2

- Write a generic optimization formulation of the stochastic market-clearing problem.
- Derive the corresponding KKT conditions of the generic formulation.
- Using KKT conditions, prove under which conditions it holds for each transmission node: *DA nodal price = expected RT nodal price*.

Guide

For the first step, check the following paper:

J. M. Morales, A. J. Conejo, K. Liu, and J. Zhong, "Pricing electricity in pools with wind producers," *IEEE Transactions on Power Systems*, vol. 27, no.3, pp. 1366-1376, 2012.

Stochastic market clearing: **equilibrium**

Similar to the deterministic market design, it is straightforward to develop an “equivalent” stochastic equilibrium model as a stochastic market-clearing mechanism.

Stochastic market clearing: equilibrium

Each conventional generator:

Maximize **expected** profit
subject to:
Production limits (in DA and RT)

Each wind generator:

Maximize **expected** profit
subject to:
Production limits (in DA and RT)

Transmission operator:

Maximize **expected** profit
subject to:
Network limits (in DA and RT)

Each load (inelastic):

Minimize **expected** cost
subject to:
Load shedding limits (in RT)

DA price-setter problem

**RT price-setter problem, one
per scenario**

Exercise 3

The available **stochastic market-clearing models** in the literature (e.g., the one explained in this lecture) are theoretical models. To the best of the lecturer's knowledge, there is no real-world electricity market that uses a stochastic clearing model (except Swiss reserve market [1]).

- Why? What are shortcomings of stochastic market-clearing models in practice and by design?

Guide:

- How straightforward is it to implement those models in practice (e.g., in terms of acquiring and processing probabilistic data needed for stochastic clearing)?
 - Is the *merit order dispatch* (i.e., scheduling generators from the cheapest one to the most expensive one) ensured in DA? Is there any possibility that a generator is paid in DA at a price lower than its offer price?
 - Is the “cost recovery” property ensured for each individual scenario?
- Explore whether there is any mechanism in practice to improve the performance of the deterministic market-clearing model (i.e., to get solutions from the deterministic model as close as those to be obtained from the stochastic one)?

[1] F. Abbaspourtorbati and M. Zima, “The Swiss reserve market: Stochastic programming in practice,” *IEEE Transactions on Power Systems*, vol. 31, no. 2, pp. 1188-1194, 2016.

Course project (step 2)

Step 2: Solving the problem under uncertainty (either option 1 or 2)

Option 1: Scenario-based stochastic programming

- Generate >100 scenarios to model the uncertain parameter (e.g., wind power)
- Pick a part of scenarios (say 20% of scenarios, either arbitrarily or the most representative ones) as training (in-sample) scenarios and solve the stochastic problem
- Using the rest of scenarios as testing data, carry out an out-of-sample analysis
- Compare the value of objective function obtained in both analyses above in terms of expected value and standard deviation
- Analyze how the selection method and the number of in-sample scenarios impact the out-of-sample results

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Discussion

What is an out-of-sample analysis?

Thanks for your attention!

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